

Neural Feedback Optimization Theory (NFOT): The Write-Back Gap as a Trial-Level Temporal Variable in Closed-Loop Learning Systems

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Note on versions. This is v3 of this preprint. Previous versions (v1, v2) contained a specific numerical claim—a characteristic timescale of approximately 28.42 ms—derived from a simulation whose assumptions could not be independently verified, together with a derived functional form (the Lorentzian efficiency law). Both have been removed. The present version states the hypothesis and a protocol for testing it, and does not claim a numerical constant or a derived kernel. Readers interested in the history of the framework should consult the previous versions.

Abstract

Closed-loop neuroadaptive systems—including brain-computer interfaces (BCIs), neuro-feedback systems, and adaptive learning technologies—detect a biological or behavioral event, compute an intervention, and deliver feedback. The elapsed time between the event and the delivered feedback is usually treated as an engineering property to be minimized. This paper argues that it should be treated as a scientific variable. I define the *Write-Back Gap* L as the achieved, trial-level interval between a prespecified biological event t_0 and the completion of the corresponding feedback t_{fb} , measured on matched pairs. I argue this differs from ordinary latency in four specific ways, and that the *distribution* of L —not only its mean—may carry information about learning outcomes.

The paper states a falsifiable predictive hypothesis and a stronger causal hypothesis, presents a preregistered experimental protocol (PTSP) for testing both, and specifies in advance what results would disconfirm them. It also surveys publicly available closed-loop datasets to determine whether the hypothesis can currently be tested. No functional form is assumed, no numerical time constant is claimed, and no simulation output is presented as a biological finding. Whether the central prediction survives is an empirical question.

1 Introduction

Closed-loop systems that adapt to a user’s biological or behavioral state share a common structure: a target event occurs, the system detects it, computes a response, and delivers feedback. The time this takes is normally treated as an engineering cost. Decoders are benchmarked on latency; architectures are optimized to reduce it; a faster pipeline is assumed to be a better one.

This paper argues that this framing misses a scientific question. Between the target event and the delivered feedback, the biological state that determines whether learning occurs may change. Synaptic eligibility traces decay. Neuromodulatory signals rise and fall. The brain does not wait for the pipeline to finish. If the temporal relationship between an event and its feedback affects learning, then that relationship is a variable worth measuring—not merely an engineering delay to be minimized.

The paper does four things. Section 2 defines the variable, called the Write-Back Gap, and distinguishes it from ordinary latency. Section 3 states the hypothesis under which it matters and specifies what would falsify it. Section 4 presents the strongest testable prediction. Section 5 surveys public datasets and reports whether the hypothesis can currently be tested. Section 6 presents a protocol for testing it.

2 The Write-Back Gap

2.1 Definition

Let t_0 denote the instant at which a prespecified target biological state transition occurs. Depending on the system, t_0 may correspond to a pre-/postsynaptic spike relationship, a decodable transition in a neural population trajectory, a behavioral or physiological event, or a crossing of a prespecified phase ϕ^* of an ongoing electrophysiological oscillation.

Let t_{fb} denote the instant at which the corresponding adaptive feedback is fully delivered to the biological substrate. The Write-Back Gap is

$$L \equiv t_{fb} - t_0, \quad L \geq 0. \quad (1)$$

The total gap decomposes as

$$L = L_{acq} + L_{filter} + L_{decode} + L_{compute} + L_{network} + L_{act} + L_{other}. \quad (2)$$

Each component can be instrumented separately, while the biological experiment operates on the achieved end-to-end value.

2.2 Why This Is Not Simply Latency

The distinction is not terminological. It consists of four commitments.

1. **Relative to what.** L is measured relative to a *biologically defined* event t_0 , not an arbitrary system timestamp. The choice of t_0 is a scientific commitment and must be made independently of the observed learning outcome.
2. **Measured at which endpoint.** L terminates at the completion of feedback delivery to the substrate, not at the moment feedback is enqueued or dispatched. Software-side timestamps and delivered-feedback timestamps can differ substantially.
3. **Correspondence.** L is defined on matched (t_0, t_{fb}) pairs. A mean latency over a session is not the same object as a per-trial gap between a specific event and the feedback generated in response to it.
4. **Treatment.** Conventional latency analysis treats latency as an engineering property to be minimized. Here L is a variable whose relationship to learning is an empirical question, whose distribution may itself be predictive, and whose connection to biological eligibility is a hypothesis rather than an assumption.

If these four commitments are not upheld, L collapses into ordinary system latency and the contribution of this paper disappears.

3 The Core Hypothesis and Its Falsification

3.1 The Hypothesis

Predictive hypothesis. After the target event and feedback modality are prespecified, the achieved trial-level Write-Back Gap L contains reproducible out-of-sample information about learning-related outcomes under at least some closed-loop conditions.

Causal hypothesis (stronger, tested separately). Manipulating L while holding relevant feedback properties constant changes the learning-related outcome.

The predictive and causal claims are not interchangeable. The predictive claim can be supported by observational data; the causal claim requires controlled manipulation.

3.2 What Would Falsify It

A framework that cannot be disconfirmed by any conceivable result is not scientific. Table 1 states, in advance, how a range of possible outcomes bears on the hypothesis.

Table 1: Null-result interpretations, stated in advance.

Empirical result	Interpretation
L improves preregistered out-of-sample prediction	Supports the predictive hypothesis
L adds no reproducible predictive information after controls	Challenges the predictive hypothesis; the framework should be abandoned as a distinct contribution
Mean L does not matter, but variance or tails do	Supports the distributional prediction while weakening a mean-only reading
Manipulating L does not change the outcome under matched feedback content	Challenges the causal hypothesis

4 The Distributional Prediction

This is the strongest testable claim in the paper.

4.1 Statement

Consider two closed-loop systems A and B with

$$\mathbb{E}[L_A] = \mathbb{E}[L_B] \quad \text{but} \quad \text{Var}(L_A) \neq \text{Var}(L_B), \quad (3)$$

and with matched feedback content, modality, and task. The prediction is that learning outcomes may differ between A and B .

4.2 Why Mean Latency Alone May Be Insufficient

If learning depends on the residual eligibility available at the moment of delivery, and if eligibility decays nonlinearly, then a system with matched mean but higher variance delivers some feedback very early (while eligibility is high) and some very late (while eligibility has largely decayed). The late trials contribute disproportionately little to learning. A mean-only summary cannot capture this asymmetry. The prediction is therefore not a background observation that latency varies; it is a specific claim that the *shape* of the latency distribution carries information that the mean does not.

4.3 How to Test It

To render the prediction falsifiable, a study must:

1. prespecify the distributional summary whose effect is tested (variance, a tail quantile, or a tail-risk functional);
2. construct conditions with matched means but different distributional summaries;
3. test whether the distributional summary predicts outcome after controlling for the mean;
4. preregister the predicted direction.

Without these commitments, “distribution matters” risks becoming unfalsifiable, because almost any latency statistic can be examined after the fact.

5 Dataset Survey

To determine whether the hypothesis can currently be tested, I surveyed publicly available closed-loop datasets. I searched OpenNeuro for keywords including “neurofeedback,” “P300 speller,” and “BCI,” and examined the most relevant candidates for whether they preserved trial-level selection or feedback timing.

5.1 BCI-FIT (OpenNeuro ds007720)

BCI-FIT is a communication BCI dataset from five participants with ALS, collected with BciPy [7]. It includes calibration and copy-spelling tasks in BIDS format.

I examined the CopyPhrase task files for participant `sub-P1`, session `ses-2`, run 7. The event files are:

- `sub-P1_ses-2_task-MatrixCopyPhrase_run-7_events.tsv` — contains `onset`, `duration`, `trial_type`, `value`, `sample`. The `trial_type` column contains only `target` and `nontarget` stimulus labels. No selection or feedback events are recorded.
- `sub-P1_ses-2_task-MatrixCopyPhrase_run-7_eeg.vmrk` — contains only `Stimulus` markers. No `Response` or selection markers.

The selection events—the moments at which the system made a character decision and delivered feedback—are not present in the published BIDS files.

5.2 Chandravadia et al. (OpenNeuro ds005028)

This dataset compares P300 flashing paradigms in online typing with language models [1]. It contains raw EEG in `.dat` and `.edf` formats. No `events.tsv`, `.vmrk`, or behavioral event files are included. Event information, if preserved, is stored inside the raw BCI2000 binary files and is not directly accessible.

5.3 Finding

Neither dataset preserves trial-level selection or feedback timing in an accessible format. This is not a criticism of the researchers—both datasets are valuable and were collected with care. It is a statement about the current state of the field: **the data required to test the Write-Back Gap hypothesis is not routinely recorded or published.**

6 Proposed Experimental Validation: PTSP

6.1 Design Principle

Standardize and preregister the trigger; randomize or systematically manipulate the delay.

The target phase $\phi^* \in [0, 2\pi)$ is preregistered according to the target system. No specific value such as 180° is predicted here.

6.2 Target Event Preregistration

The definition of t_0 is a scientific commitment, not a free parameter. If researchers select the event definition producing the strongest observed relationship with learning, the hypothesis becomes unfalsifiable by construction. PTSP therefore requires that the operational definition of t_0 (signal, filter, detector, threshold, phase) be specified before outcome data are examined, justified by independent reasoning, and reported as an exploratory deviation if changed.

6.3 Latency Conditions

Three conceptual conditions:

- **Fast arm.** Low-latency pipeline, candidate range $L \in [1, 20]$ ms.
- **Slow arm.** Conventional pipeline, candidate range $L \in [50, 500]$ ms.
- **Programmable-delay arm.** A controlled delay stage samples intermediate gaps and permits direct manipulation of both latency and jitter.

These are proposed experimental regions, not biological thresholds.

6.4 Threats to Causal Identification

The central challenge is that fast and slow conditions may not deliver “the same feedback at different times.” Participants may experience them as qualitatively different events. PTSP specifies the following controls:

1. randomized trial-level delay assignment;
2. identical feedback content across conditions;
3. identical feedback duration and intensity;
4. blinding where possible;
5. counterbalancing of order, fatigue, and practice;
6. analysis on achieved latency L_i , not nominal condition labels;
7. independent manipulation checks for perceived contingency and temporal predictability;
8. baseline block before the manipulation;
9. primary outcome measured after the intervention, not during it;
10. where feasible, temporal predictability manipulated as a separate factor with matched mean.

6.5 Power and Trial Allocation

The delay range should be sampled approximately uniformly on a log scale, with sufficient trials in the tail to distinguish competing distributions. A design concentrated near zero delay has almost no power to discriminate heavy-tailed from light-tailed alternatives. The primary analysis should specify a minimum effect size of theoretical interest in advance, so that a null result is interpretable as evidence rather than inconclusive.

6.6 Primary Outcome and Candidate Models

The primary dependent variable should be defined independently of the latency manipulation. A useful formulation is

$$Y_i = f(L_i, \mathbf{z}_i; \theta) + \epsilon_i, \quad (4)$$

where Y_i is the learning-related outcome, L_i the achieved gap, $\mathbf{z}_i$ prespecified covariates, and ϵ_i unexplained variation. The critical test is whether L_i explains out-of-sample variation in Y_i after controls.

If a functional form is fit, it must compete against alternatives. Candidate forms include

$$E_{\text{exp}}(L) = e^{-L/\tau}, \quad (5)$$

$$E_{\text{Gauss}}(L) = e^{-(L/\sigma)^2}, \quad (6)$$

$$E_{\text{pow}}(L) = \left(1 + \frac{L}{\tau}\right)^{-\alpha}, \quad (7)$$

$$E_{\text{L}}(L) = \frac{c}{L^2 + c}, \quad c > 0. \quad (8)$$

No candidate has theoretical priority. The Lorentzian form is included because its heavy tail is one plausible shape, not because it is derived from biology. Model comparison should use held-out or cross-validated predictive criteria, and all fitted parameters should be reported with uncertainty.

7 Simulations and Their Proper Interpretation

Simulations of the proposed closed-loop structure can illustrate what the framework implies under specified assumptions. They cannot establish biological constants.

Any simulation reported in this work should state:

1. every parameter value and its source (specified, estimated, or arbitrary);
2. the sampling rate, buffer size, decay function, and random seed;
3. what happens to the output when each parameter is varied;
4. and, explicitly, that the output is a property of the simulation, not a measurement of the brain.

If a simulation generates synthetic data and then fits a candidate model to that data, the resulting fit reflects the assumptions used to generate the data, not the empirical adequacy of the model. Such exercises can be useful for checking that an inference procedure recovers known ground truth, but they are not evidence for a biological claim.

8 Scope and Limitations

This paper proposes a measurement and a testable prediction. It does not propose a biological mechanism, does not derive a functional form, and does not present simulation output as evidence.

- The predictive hypothesis and the distributional prediction are the only scientific claims. If they fail, the contribution reduces to a measurement convention.
- The causal hypothesis requires controlled manipulation and has not been tested here.
- No kernel is derived from biology. If a kernel is fit empirically, it must compete against alternatives and be validated out of sample.
- No numerical time constant is claimed. Any characteristic scale that appears in a candidate model is a fitted parameter, not a biological constant.
- Biological eligibility traces may be multi-timescale, non-exponential, and state-dependent. The simple exponential form used for illustration is not a biological model.
- If potentiation- and depression-associated eligibility traces differ in their temporal persistence, this would predict an asymmetry in the effective Write-Back Gap. Testing this asymmetry is future work; no specific time constants are claimed here.
- Software implementations of these ideas are engineering artifacts, not evidence for the framework.
- Physiological state is multidimensional; no single measure should be assumed to summarize it.
- Feedback modality may interact with the timing effect in ways not captured by a scalar L .
- Causal identification is difficult because latency manipulation may inadvertently alter feedback quality, predictability, or workload.

9 Research Roadmap

The framework can be evaluated as a sequence of increasingly demanding tests.

1. **Measurement validation.** Establish test–retest reliability of L at trial resolution.
2. **Distributional characterization.** Quantify mean, variance, jitter, and tail behavior.
3. **Synthetic validation.** Stress-test the identification procedure using synthetic datasets with known ground-truth kernels, to confirm that the analysis can recover a known answer before it is applied to real data.
4. **Controlled manipulation.** Vary L while holding feedback content constant, following PTSP.
5. **Model competition.** Compare candidate functions on held-out data with adequate tail sampling.
6. **Distributional test.** Test whether variance or tail statistics predict outcome after controlling for the mean.
7. **Independent replication.** Estimate parameters in independent samples and contexts.

8. **Engineering translation.** Only after biological timing effects are supported, evaluate whether architectures improving the relevant distribution produce measurable benefits.

10 Conclusion

The Write-Back Gap is proposed as a common temporal variable for studying closed-loop adaptive systems. The central idea is that feedback does not occur in a temporal vacuum: between a target biological event and the delivery of an adaptive intervention, the biological state relevant to learning may change.

The strongest claim in this paper is not that a particular latency is optimal, nor that any specific functional form is correct. It is the more general and testable proposition that the temporal distribution of feedback may contain information about learning outcomes that is lost when systems are described only by task-level performance or average latency.

A survey of public datasets suggests that the data required to test this hypothesis does not currently exist in accessible form. This is not a negative result about the hypothesis. It is a statement about the state of the field. The next step is to record the events.

Whether the proposition survives is an empirical question. The contribution of this paper is to state it precisely enough to be tested, to document that the data does not yet exist, and to specify the protocol by which it could be falsified.

A closed-loop learning system should be evaluated not merely by what feedback it provides, but by when, how consistently, and under what biological conditions that feedback is received.

References

- [1] N. Chandravadia et al., “Comparing P300 Flashing paradigms in online typing with language models,” OpenNeuro, 2024. doi:10.18112/openneuro.ds005028.v1.0.0.
- [2] N. Frémaux and W. Gerstner, “Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules,” *Frontiers in Neural Circuits*, vol. 9, p. 85, 2016.
- [3] W. Gerstner, M. Lehmann, V. Liakoni, D. Corneil, and J. Brea, “Eligibility traces and plasticity on behavioral time scales,” *Frontiers in Neural Circuits*, vol. 12, p. 53, 2018.
- [4] J. M. Horschig, W. Oosterheert, R. Oostenveld, and O. Jensen, “Modulation of posterior alpha activity by spatial attention allows for controlling a continuous brain–computer interface,” *Brain Topography*, 2015.
- [5] E. M. Izhikevich, “Solving the distal reward problem through linkage of STDP and dopamine signaling,” *Cerebral Cortex*, vol. 17, no. 10, pp. 2443–2452, 2007.
- [6] H. S. Nogay and T. C. Akinci, “Single-Trial Real-Time-Compatible P300 Latency Estimation Using a Deployment-Consistent Deep Learning Regression Framework,” *IEEE Access*, 2026.
- [7] B. Peters et al., “BCI-FIT: Effects of cBCI customization on performance,” 11th International BCI Meeting, Banff, 2025.
- [8] J. Tidare, E. Åstrand, and M. Ekström, “Evaluation of closed-loop feedback system delay: a time-critical perspective for neurofeedback training,” *BIOSIGNALS*, 2018.